

Scripts and Commands Reference

Everything used in the project, with an explanation of each command and script.

1. Setup Commands

```
sudo apt update
sudo apt upgrade -y
sudo apt install -y build-essential git bc bison flex libssl-dev libncurses5-dev libelf-dev
sudo apt install -y systemd systemd-sysv
```

The build tools are needed only for kernel rebuilding (future scope). Boot analysis itself needs only systemd.

2. Analysis and Optimization Commands

#	Command	Purpose
1	systemd-analyze	Total boot time (kernel + userspace)
2	systemd-analyze blame	Units ordered by startup time
3	systemd-analyze critical-chain	Chain of units that directly delays boot completion
4	dmesg -T	Kernel messages with readable timestamps
5	free -h	Memory usage in human-readable form
6	vmstat 1	CPU and system activity every second
7	make menuconfig	Open the kernel configuration menu
8	make -j4	Build the kernel using 4 parallel jobs
9	sudo systemctl disable <service>	Stop a service from starting at boot
10	sudo systemctl stop <service>	Stop a running service immediately
11	sudo systemctl enable <service>	Undo a disable (restore the service)

3. boot_analysis.sh

```
#!/bin/bash
echo "Embedded Linux Boot Analysis"
echo "-----"
echo "1. Total Boot Time:"
systemd-analyze
echo ""
echo "2. Slowest Services:"
systemd-analyze blame | head -10
echo ""
echo "3. Critical Boot Chain:"
systemd-analyze critical-chain
echo ""
echo "4. Memory Usage:"
free -h
echo ""
echo "5. Recent Kernel Messages:"
dmesg | tail -20
```

Run it with:

```
chmod +x boot_analysis.sh
./boot_analysis.sh
```

Line-by-line explanation

Line	Explanation
<code>#!/bin/bash</code>	Shebang: run the script with the Bash shell
<code>echo "..."</code>	Prints a title or heading
<code>systemd-analyze</code>	Shows total time taken to boot
<code>systemd-analyze blame head -10</code>	Lists services by startup time; pipes the output and head -10 keeps the first 10 lines
<code>systemd-analyze critical-chain</code>	Shows the chain of units that decides when boot completes
<code>free -h</code>	Memory usage in MB/GB
<code>dmesg tail -20</code>	Last 20 kernel log lines

4. optimize_services.sh

```
#!/bin/bash
echo "Starting service optimization..."

services=("bluetooth" "cups" "avahi-daemon")

for service in "${services[@]}; do
    echo "Disabling $service ..."
    sudo systemctl disable "$service"
done

echo "Optimization complete."
echo "Please reboot and check boot time using:"
echo "systemd-analyze"
```

Run it with:

```
nano optimize_services.sh      # create and paste the script
chmod +x optimize_services.sh
./optimize_services.sh
sudo reboot
```

Line-by-line explanation

Line	Explanation
#!/bin/bash	Run with Bash
echo "Starting service optimization..."	Prints a start message
services=("bluetooth" "cups" "avahi-daemon")	Array holding the service names to disable
for service in "\${services[@]}; do	Loops over every name in the array; the semicolon before do is required
echo "Disabling \$service ..."	Prints which service is being processed
sudo systemctl disable "\$service"	Prevents the service from starting automatically at boot
done	Ends the loop
echo "Optimization complete." ...	Completion message and reminder to reboot and run systemd-analyze

Important notes

- `disable` takes effect at the next boot. To stop a service right now as well, use `sudo systemctl disable --now <service>`.
- If a listed service is not installed, `systemctl` prints a warning and the script continues.
- Only disable services your system does not need. Do not disable network services if your project depends on networking.
- To restore a service: `sudo systemctl enable <service>`.

5. Measurement Checklist

Step	Command	Record
Before	systemd-analyze	Startup finished / graphical.target time
Before	systemd-analyze blame head -10	Top 10 slowest units
Optimize	./optimize_services.sh, then sudo reboot	-
After	systemd-analyze	Startup finished / graphical.target time
After	systemd-analyze blame head -10	Top 10 slowest units